

UBERABA, Brazil

WMO: 835760

Lat: 19.765S

Lon: 47.965W

Elev: 809

StdP: 91.97

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	11.1	13.2	0.2	4.2	25.5	2.0	4.8	25.1	8.6	22.7	7.8	22.9	2.9	240	0.327

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	10.6	34.0	18.3	33.0	18.8	32.0	19.2	23.3	28.0	22.8	27.7	22.4	27.3	3.8	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	18.6	24.3	21.8	18.2	24.0	21.2	17.5	23.6	73.8	27.9	71.9	28.1	70.4	27.3	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.0	7.0	6.2	DB	7.8	35.4	2.6	1.3	5.9	36.4	4.4	37.1	2.9	37.9	1.0	38.9	(4)
(5)			WB	5.7	24.8	2.6	1.6	3.9	25.9	2.4	26.8	1.0	27.7	-0.9	28.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	23.8	24.8	25.2	24.8	23.9	21.9	21.0	21.2	23.0	24.9	25.4	24.6	24.7	(6)	
	DBStd	2.54	1.57	1.30	1.44	1.86	2.15	2.29	2.46	2.22	2.61	2.50	1.84	1.44	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	23	0	0	0	1	3	7	9	2	1	0	0	0	(9)	
	CDD10.0	5028	459	427	458	416	368	330	348	404	446	478	437	457	(10)	
	CDD18.3	2009	201	193	199	166	113	87	99	148	197	219	187	199	(11)	
	CDH23.3	14923	1336	1389	1277	1131	687	551	708	1284	1927	2047	1315	1271	(12)	
(13)	CDH26.7	5094	447	486	408	327	135	84	143	442	865	909	449	399	(13)	
(14)	Wind	WSAvg	3.2	3.1	2.9	2.9	2.9	3.1	3.3	3.5	3.7	3.5	3.5	3.3	(14)	
(15) Precipitation	PrecAvg	1484	289	222	197	71	49	17	11	15	57	107	189	253	(15)	
	PrecMax	1824	569	345	371	155	122	85	64	59	130	242	359	364	(16)	
	PrecMin	1092	97	103	106	0	2	0	0	0	3	42	108	139	(17)	
	PrecStd	192	120	82	68	41	32	23	16	16	36	51	63	53	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.9	33.0	32.2	32.2	30.2	29.8	30.2	33.2	35.1	36.1	33.8	32.2	(19)	
		MCWB	20.8	21.5	21.1	21.5	18.7	17.1	16.5	16.7	17.3	18.1	20.1	20.9	(20)	
	2%	DB	31.8	31.9	31.1	30.9	29.1	28.4	29.1	31.8	34.0	34.2	32.0	31.1	(21)	
		MCWB	20.8	21.0	21.1	20.7	18.7	17.1	16.5	16.5	17.5	18.8	20.2	21.0	(22)	
	5%	DB	30.8	30.9	30.1	29.8	28.1	27.8	28.2	30.5	32.9	33.0	30.9	30.1	(23)	
		MCWB	20.8	21.0	21.1	20.2	18.3	17.1	16.3	16.0	17.3	18.9	20.4	21.2	(24)	
	10%	DB	29.2	29.9	29.1	28.8	27.1	26.8	27.1	29.1	31.2	31.8	29.2	29.1	(25)	
		MCWB	21.2	21.0	21.2	20.0	18.0	16.9	16.1	15.8	17.3	19.0	20.4	21.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.7	23.8	23.2	23.3	21.6	20.2	19.6	19.3	21.7	22.6	23.6	24.0	(27)	
		MCDB	27.6	29.1	28.4	28.6	26.4	25.3	25.2	27.6	29.4	30.0	27.9	27.6	(28)	
	2%	WB	23.0	23.0	22.7	22.2	20.7	19.2	18.4	18.3	20.7	21.7	22.5	22.9	(29)	
		MCDB	27.2	28.3	28.1	27.2	25.4	24.9	25.2	26.3	27.5	28.1	27.3	27.4	(30)	
	5%	WB	22.5	22.5	22.2	21.6	20.1	18.6	17.6	17.6	20.0	21.1	21.9	22.4	(31)	
		MCDB	26.8	27.7	27.6	26.5	25.0	24.5	25.2	25.6	26.6	26.8	26.8	27.0	(32)	
	10%	WB	22.1	22.1	21.8	21.1	19.2	17.9	16.9	17.0	19.2	20.6	21.4	22.0	(33)	
		MCDB	26.4	27.1	26.9	25.8	24.2	23.8	24.6	25.4	25.8	26.4	26.0	26.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	9.2	8.9	9.6	9.9	10.7	11.3	11.9	11.6	10.6	9.3	8.8	(35)	
		MCDBR	10.5	10.7	10.2	10.6	11.1	11.9	12.3	13.1	12.9	12.3	11.2	10.3	(36)	
	5% WB	MCWBR	3.5	3.5	3.4	3.7	4.1	4.5	4.4	4.3	4.2	3.7	3.7	3.4	(37)	
		MCWBR	8.7	9.4	9.0	9.2	9.6	10.3	11.3	11.7	10.9	10.3	9.0	8.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.384	0.383	0.371	0.356	0.336	0.335	0.344	0.373	0.469	0.439	0.395	0.390	(40)	
		taud	2.471	2.458	2.510	2.521	2.534	2.539	2.485	2.378	2.114	2.263	2.405	2.443	(41)	
	Ebn at Noon		961	950	937	913	895	877	878	881	832	887	943	955	(42)	
		Edh at Noon	119	119	109	100	92	88	95	114	159	143	126	122	(43)	
(44) All-Sky Solar Radiation	RadAvg		5.65	5.91	5.37	5.12	4.48	4.23	4.50	5.27	5.57	5.84	5.62	5.73	(44)	
	RadStd		0.68	0.47	0.47	0.32	0.23	0.21	0.25	0.30	0.51	0.43	0.29	0.38	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page